

Mitigating Catastrophic Forgetting in EN-DE Idiom Translation via Two-Stage Fine-Tuning and Metric-Aware Evaluation

Evan Powell

University of California, Berkeley
evan.powell@berkeley.edu

Michael Strommer

University of California, Berkeley
michael.strommer@berkeley.edu

Abstract

Neural machine translation (NMT) systems achieve strong performance on standard benchmarks but remain brittle when handling idiomatic expressions, which require non-compositional understanding. We propose a two-stage fine-tuning approach that first specializes models on idiomatic data and then re-aligns them with general-domain translation. Our method improves idiom correctness from 0.44 to 0.58 (+14pp) in human evaluation while maintaining strong general-domain performance. Crucially, we show that conclusions about catastrophic forgetting depend on the evaluation metric: while BLEU suggests degradation, COMET and human judgments indicate preserved or improved semantic adequacy. These findings highlight both an effective training strategy for idiomatic translation and the importance of metric choice in evaluating NMT systems.

1 Introduction

Neural machine translation (NMT) has made significant progress in recent years, achieving near-human performance on many standard benchmarks. However, these systems remain brittle when handling idiomatic expressions, where meaning cannot be derived compositionally from individual words. For example, the English phrase “kick the bucket” cannot be translated literally into German without losing its intended meaning. Such failures highlight a fundamental limitation: while modern models excel at surface-level fluency, they often struggle to capture deeper semantic meaning.

This limitation is particularly important because idiomatic language is pervasive in real-world communication, including conversational text, journalism, and literature. Errors in idiom translation can lead to outputs that are fluent but semantically incorrect, reducing trust in machine translation systems.

A straightforward approach is to fine-tune models on idiom-specific datasets. However, prior work has shown that such specialization often leads to catastrophic forgetting, degrading performance on general-domain translation tasks. This creates a fundamental tension between specialization and generalization.

In this work, we propose a two-stage fine-tuning approach that explicitly addresses this trade-off. In the first stage, the model is specialized on idiomatic data to learn non-compositional mappings. In the second stage, the model is re-aligned using general-domain data, optionally combined with a small proportion of idiomatic examples. We further explore architectural interventions such as encoder freezing to stabilize learning.

We also compare against parameter-efficient LoRA variants and prompt-based approaches using an instruction-tuned FLAN-T5 model.

In addition to modeling contributions, we investigate the role of evaluation metrics. Traditional metrics such as BLEU may fail to capture semantic improvements in idiom translation. We therefore incorporate COMET and conduct a blinded human evaluation to provide a more comprehensive assessment.

Our contributions are threefold: (1) we introduce a two-stage fine-tuning strategy for idiom translation, (2) we empirically demonstrate how catastrophic forgetting can be mitigated, and (3) we show that evaluation conclusions depend strongly on the choice of metric.

2 Background

Idiomatic expressions remain a persistent challenge for machine translation due to their non-compositional nature, where meaning cannot be inferred from individual words. Prior work has shown that neural machine translation systems struggle to correctly translate idioms, often de-

faulting to literal renderings that preserve fluency but distort meaning [Fadaee et al., 2018]. More broadly, recent studies on verbal multiword expressions demonstrate that such non-literal constructions systematically degrade translation quality across languages, suggesting a general weakness of MT systems in handling non-compositional language [Liu et al., 2025].

A natural approach to address this limitation is to fine-tune models on specialized data. However, this introduces the risk of catastrophic forgetting, where optimizing for a target domain degrades performance on previously learned tasks [Kirkpatrick et al., 2017]. In the context of machine translation, recent work has highlighted a related phenomenon, showing that fine-tuning can improve translation quality on specific tasks while simultaneously degrading other capabilities—a tension referred to as the *fine-tuning paradox* [Stap et al., 2024]. These findings suggest that improving specialized capabilities, such as idiom translation, may come at the cost of general-domain performance.

Modern neural machine translation systems are built on Transformer architectures, which rely on self-attention to model long-range dependencies and have become the dominant paradigm for sequence-to-sequence tasks [Vaswani et al., 2017]. While these models achieve strong overall performance, their reliance on statistical patterns in training data makes them particularly vulnerable to errors on rare or non-compositional constructions.

An alternative to full fine-tuning is parameter-efficient fine-tuning (PEFT), which adapts pre-trained models by updating only a small subset of parameters. Among these methods, LoRA (Low-Rank Adaptation) has emerged as a widely used approach for efficient adaptation, making it possible to modify model behavior with substantially lower computational cost than full fine-tuning [Hu et al., 2022].

Finally, evaluation plays a critical role in assessing these trade-offs. Traditional metrics such as BLEU and chrF measure surface-level overlap between predicted and reference translations [Papineni et al., 2002, Popović, 2015], but may fail to capture improvements in semantic adequacy. In contrast, learned evaluation metrics such as COMET explicitly model semantic similarity and have been shown to correlate more strongly with human judgments [Rei et al., 2020].

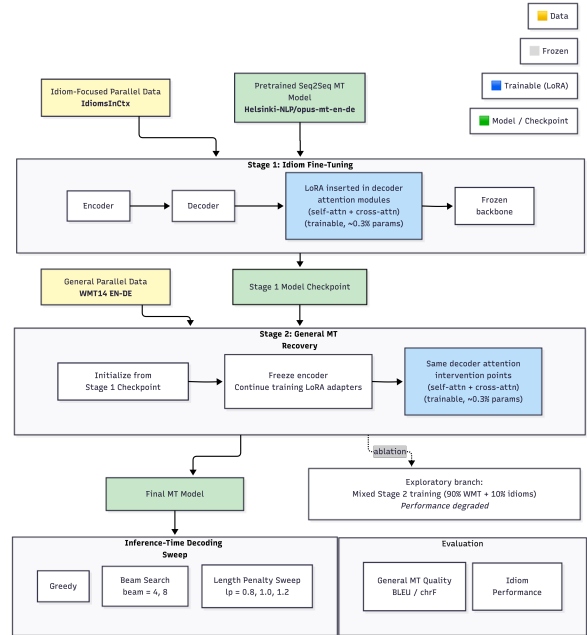


Figure 1: Two-Stage Fine-Tuning Pipeline

Our work builds on these insights by studying idiom translation as a specialized capability and proposing a two-stage fine-tuning strategy to balance specialization and generalization. Importantly, we also examine how evaluation metrics influence conclusions about catastrophic forgetting.

3 Methods

3.1 Problem Setup

We consider the task of English-to-German machine translation with a particular focus on idiomatic expressions. Given a source sentence x , the goal is to produce a translation y that is both fluent and semantically faithful, including correct idiomatic rendering when applicable.

We evaluate models on two complementary datasets: (1) an idiom-focused dataset (*IdiomsInCtx*) containing 1,000 sentence pairs with idiomatic expressions, and (2) a general-domain dataset (*WMT14 EN-DE*). This dual evaluation setup is critical, as it allows us to explicitly measure the trade-off between specialization (idioms) and generalization (standard translation).

3.2 Two-Stage Fine-Tuning

Our approach consists of two sequential fine-tuning stages designed to decouple the learning of idiomatic knowledge from general translation ability.

In the first stage, the base model is fine-tuned

exclusively on idiomatic data. This stage encourages the model to override its default tendency toward literal translation and instead learn non-compositional mappings specific to idiomatic expressions. Empirically, we find that this stage substantially improves idiom translation quality, but also shifts the model distribution away from general-domain patterns.

In the second stage, the model is further fine-tuned on general-domain data to restore overall translation performance. We explore two variants of this stage: (1) pure general-domain fine-tuning and (2) mixed training, where a small proportion of idiomatic examples (10%) is retained. The latter is motivated by the hypothesis that continued exposure to idioms during re-alignment can prevent the model from “unlearning” specialized knowledge.

Additionally, we investigate freezing the encoder during Stage 2. This intervention is intended to preserve general linguistic representations learned during pretraining while allowing the decoder to adapt to domain-specific output distributions.

Overall, this staged approach is motivated by the intuition that idiomatic knowledge can be learned in isolation and then integrated into a broader translation capability without being overwritten.

3.3 Parameter-Efficient Fine-Tuning

In addition to full fine-tuning, we experiment with parameter-efficient approaches using Low-Rank Adaptation (LoRA). LoRA adapts a pre-trained model by injecting low-rank trainable matrices into existing weight layers, while keeping the original model parameters frozen. This allows models to be fine-tuned with significantly fewer trainable parameters—in our setup, only approximately 0.3% of the total model parameters are updated.

A key hyperparameter in LoRA is the rank r of the low-rank decomposition, which controls the capacity of the adaptation. Larger values of r allow the model to learn more expressive updates, while smaller values constrain the adaptation and may limit overfitting. In our experiments, we evaluate multiple configurations ($r \in \{4, 8, 16\}$) to study how adaptation capacity affects the trade-off between idiom specialization and general-domain performance.

3.4 Baselines and Prompt-Based Approaches

We compare the proposed approach against several baselines, including the base model without fine-tuning (*Helsinki-NLP/opus-mt-en-de*) and an idiom-only fine-tuned model. The latter serves as a strong specialization-focused comparison point but is expected to suffer from catastrophic forgetting.

To evaluate whether idiomatic translation can be achieved without parameter updates, we also include prompt-based methods using a separate instruction-tuned *FLAN-T5* model. We test zero-shot and few-shot prompting (3-shot and 5-shot) to assess the effectiveness of in-context learning for idiomatic translation.

3.5 Evaluation Metrics

We evaluate models using a combination of automatic and human metrics. Automatic metrics include BLEU, which measures n-gram overlap, chrF, which captures character-level similarity, and COMET, a learned metric designed to better reflect semantic adequacy.

Importantly, these metrics capture different aspects of translation quality. BLEU and chrF are sensitive to lexical overlap, while COMET is trained to approximate human judgments of semantic fidelity. This distinction becomes central to our analysis, as we observe systematic differences in conclusions depending on the metric used.

In addition, we conduct a blinded human evaluation on a balanced set of 50 sentences (25 idioms and 25 WMT). One annotator rated translations based on fluency, adequacy, and idiom correctness, following an ACL-style evaluation protocol.

3.6 Experimental Design and Iteration

Our experimental process is iterative and driven by observed failure modes. We begin with a baseline model and progressively introduce modifications, including idiom-only fine-tuning, two-stage training, encoder freezing, mixed-data training, and LoRA-based adaptations.

Each model is evaluated on both idiom and general-domain datasets, enabling direct comparison across specialization and generalization dimensions. This design allows us not only to measure performance improvements, but also to understand how different training strategies shift the balance between competing objectives.

Model	Idioms			WMT		
	BLEU	chrF	COMET	BLEU	chrF	COMET
Baseline	39.65	60.79	0.733	27.57	58.43	0.833
Idiom-only	44.16	64.44	0.763	26.54	58.23	0.863
Two-stage	43.16	63.34	0.761	27.18	58.34	0.847
Mixed-data	43.49	62.92	0.741	26.92	58.10	0.852
LoRA-4	41.85	61.84	0.729	27.57	58.49	0.833
LoRA-8	42.18	61.76	0.732	27.48	58.46	0.842
LoRA-16	41.72	61.45	0.733	27.60	58.46	0.847
Prompt (0-shot)	8.13	33.14	0.381	7.62	35.54	0.474

Table 1: Automatic evaluation by split. Idiom-only fine-tuning achieves the strongest performance on idioms, while two-stage fine-tuning provides a strong balance between specialization and generalization. Notably, metric-dependent differences emerge: idiom-only fine-tuning decreases BLEU on WMT but improves COMET, highlighting divergence between lexical and semantic evaluation.

4 Results and Discussion

4.1 Automatic Evaluation: Specialization vs Generalization

The automatic evaluation results (Table 1) reveal a clear and nuanced trade-off between idiom specialization and general-domain performance. On the idiom dataset, idiom-only fine-tuning achieves the strongest results across all metrics, improving BLEU from 39.65 to 44.16 and COMET from 0.733 to 0.763. This confirms that targeted supervision enables the model to learn non-compositional mappings effectively.

However, the effect of this specialization on general-domain translation depends strongly on the evaluation metric. On the WMT dataset, idiom-only fine-tuning reduces BLEU from 27.57 to 26.54, suggesting degradation under overlap-based metrics. In contrast, COMET increases from 0.833 to 0.863, indicating improved semantic adequacy despite the apparent loss in lexical similarity.

This divergence highlights a key finding: **catastrophic forgetting is not a binary phenomenon, but rather depends on how performance is measured**. Under BLEU, idiom-only fine-tuning appears harmful, while under COMET it preserves—and even improves—general translation quality.

4.2 Two-Stage Fine-Tuning as a Balanced Strategy

The two-stage fine-tuning approach provides the most consistent balance between specialization and generalization. On the idiom dataset, it achieves near-peak performance (BLEU 43.16, COMET 0.761), only slightly below the idiom-

only model. On the WMT dataset, it recovers much of the BLEU loss (27.18) while maintaining strong COMET performance (0.847).

Compared to idiom-only fine-tuning, the two-stage approach sacrifices a small amount of idiom performance in exchange for improved general-domain stability. This trade-off positions it as the most robust overall strategy, particularly when evaluated across both datasets and metrics.

Notably, LoRA achieves competitive performance while updating only 0.3% of model parameters, suggesting that idiomatic knowledge can be incorporated efficiently without full model adaptation.

Mixed-data training yields similar but slightly less consistent results, suggesting that explicitly structuring the training process into separate stages provides a cleaner separation between learning and consolidation phases.

4.3 Human Evaluation and Metric Alignment

Human evaluation results (Table 2) further clarify these findings. The two-stage models the highest idiom correctness rate (0.58), outperforming both the baseline (0.44) and the idiom-only model (0.54). They also achieve the highest adequacy score (1.84), indicating improved semantic fidelity.

Notably, these results align closely with COMET rather than BLEU. While BLEU suggests that idiom-only fine-tuning degrades general performance, human judgments indicate that semantic quality is preserved or improved. This reinforces our central claim that **semantic metrics are better suited for evaluating idiomatic translation than overlap-based metrics**.

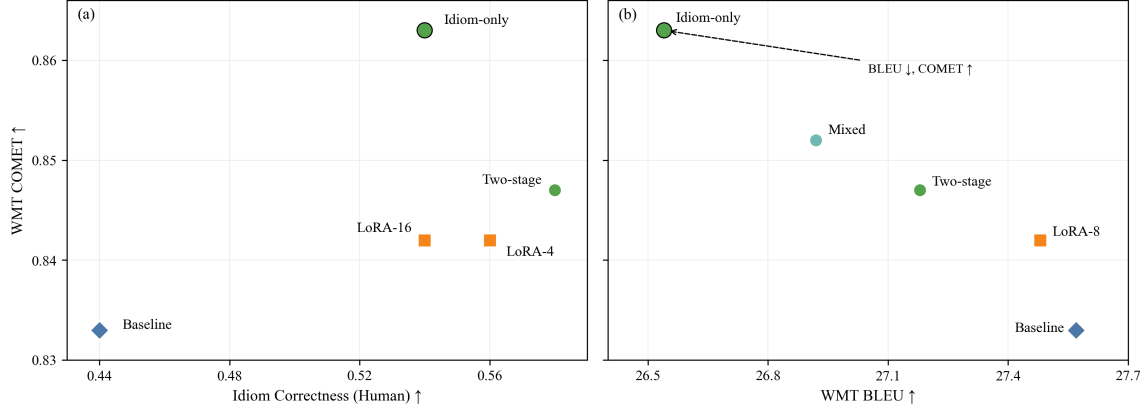


Figure 2: Metric-dependent trade-offs in idiom-aware machine translation. (a) Trade-off between idiom correctness (human evaluation) and general-domain semantic quality (WMT COMET). Two-stage fine-tuning lies on the Pareto frontier, achieving the best balance between specialization and generalization. (b) Metric disagreement on WMT: idiom-only fine-tuning decreases BLEU but improves COMET, demonstrating that apparent catastrophic forgetting is partly an artifact of evaluation choice.

Model	Fluency	Adequacy	Idiom Correct.
Baseline	1.78	1.52	0.44
Idiom-only	1.82	1.54	0.54
LoRa-4	1.84	1.54	0.56
LoRA-8	1.80	1.58	0.58
LoRA-16	1.80	1.58	0.54
Prompt	0.68	0.44	0.02

Table 2: Human evaluation (25 idioms + 25 WMT). Two-stage fine-tuning (LoRA-4/-8/-16) achieves the highest adequacy and idiom correctness while maintaining fluency.

4.4 Prompt-Based Methods

Prompt-based approaches perform substantially worse across all metrics and datasets. Even in the best case (0-shot prompting), the model achieves an idiom correctness rate of only 0.02, indicating near-complete failure on idiomatic translation. Increasing the number of examples does not yield meaningful improvements.

These results suggest that idiomatic translation requires parameter adaptation rather than relying solely on in-context learning. While prompt tuning is effective for many tasks, it appears insufficient for capturing non-compositional linguistic phenomena.

In contrast, parameter-efficient approaches such as LoRA remain competitive with full fine-tuning methods, particularly on general-domain data, highlighting that effective adaptation does not necessarily require updating all model parameters.

While these aggregate results highlight differences across training strategies and evaluation set-

tings, they do not fully explain *why* models are ranked differently. To better understand these discrepancies, we turn to qualitative analysis of representative examples.

4.5 Qualitative Analysis

To better understand how different models handle idiomatic and general-domain translation, we examine representative examples selected using sentence-level COMET spread. We present a subset of models that capture the main trade-offs discussed in the paper, including baseline, idiom-only fine-tuning, two-stage fine-tuning, and LoRA.

Rather than presenting full examples inline, we provide detailed outputs in Appendix A.1. These examples illustrate both clear failures and more subtle differences in translation adequacy and stylistic naturalness across settings.

Taken together, these examples show that differences between models are driven by semantic correctness, preservation of meaning, and stylistic naturalness rather than surface-level similarity alone. This helps explain why semantic metrics such as COMET, and human evaluation, can lead to different conclusions than overlap-based metrics.

4.6 Error Analysis and Model Behavior

The qualitative examples above highlight several recurring failure modes that extend beyond individual sentences and appear consistently across models. Baseline models frequently produce lit-

eral translations of idioms, resulting in semantically incorrect outputs. Idiom-only fine-tuning successfully increases idiom usage, but occasionally introduces unnatural phrasing or overgeneralization.

The second stage of fine-tuning mitigates these issues by restoring fluency and stabilizing output distributions. Encoder freezing further reduces drift in general linguistic structure, while mixed-data training helps retain idiomatic knowledge.

Overall, these findings suggest that idiomatic translation is best understood as a *specialized capability* that must be carefully integrated into a general-purpose model, rather than learned implicitly through standard training data.

5 Conclusion

We address the challenge of idiom translation in neural machine translation by introducing a two-stage fine-tuning approach that separates specialization from generalization. Our results show that idiomatic competence can be substantially improved without sacrificing overall translation quality, achieving a +14 percentage point gain in idiom correctness under human evaluation.

More importantly, we demonstrate that conclusions about catastrophic forgetting depend critically on the evaluation metric. While overlap-based metrics such as BLEU suggest degradation, semantic metrics and human judgments reveal that general translation quality is largely preserved. This suggests that some instances of “forgetting” may be artifacts of evaluation rather than true losses in capability.

These findings highlight the need for both structured training strategies and metric-aware evaluation when assessing modern NMT systems. Future work could extend this approach to larger models, additional language pairs, and more advanced continual learning methods.

6 Author Contributions

This project was conducted collaboratively, with both authors contributing equally to the overall design, execution, and writing of the work.

Both authors jointly explored potential research directions, ultimately selecting idiom-aware machine translation as the focus of the project. We also conducted a shared review of related literature to ground the study in prior work on idiom trans-

lation, domain adaptation, and catastrophic forgetting.

Michael Strommer led the development of the initial modeling framework, including the baseline and two-stage fine-tuning pipelines, as well as the parameter-efficient LoRA ablation experiments. He also implemented the initial analysis pipeline, conducted early evaluation using BLEU and chrF metrics, and annotated the sentence predictions.

Evan Powell focused on refining and extending the experimental pipeline, including the integration of COMET-based evaluation and the development of additional model variants such as mT5 and FLAN-T5 for prompt-based comparisons. He also designed and implemented the automatic evaluation pipeline refinements and the human evaluation framework, including the associated analysis notebooks.

Both authors contributed extensively to interpreting results, refining experimental design, and writing the final paper. The final paper was developed iteratively with substantial input and revisions from both authors throughout all stages of the project.

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A Appendix

A.1 Qualitative Examples

Example 1: Idiomatic Expression

Example 1 (Idiom)

Source: Bend the knee, peasant! Admit that I am your rightful king!

Reference: Beuge das Knie, Bauer! Gib zu, dass ich dein rechtmäßiger König bin!

Baseline: Gib zu, dass ich dein rechtmäßiger König bin!

Two-stage: Kniebeugen, Bauer! Gib zu, dass ich dein rechtmäßiger König bin!

Idiom-only: Kniebeugen, Bauer! Gib zu, dass ich dein rechtmäßiger König bin!

LoRA r=16: Gib zu, dass ich dein rechtmäßiger König bin!

COMET: 0.799 / 0.917 / 0.917 / 0.799

This example highlights a different type of error. The baseline and LoRA outputs omit the opening imperative entirely, losing part of the source meaning. In contrast, the two-stage and idiom-only models retain more of the sentence structure, but render “bend the knee” as the literal and stylistically awkward “Kniebeugen.” Although these outputs are not fully natural, they better preserve the intent of the original sentence. This trade-off between completeness and idiomatic naturalness is reflected in the COMET scores.

Example 2: General-Domain Sentence

Example 2 (WMT)

Source: “Decisions will then be made elsewhere,” said Maier.

Reference: “Dann werden Entscheidungen an anderer Stelle getroffen”, sagte Maier.

Baseline: “Beschlüsse werden dann an anderer Stelle getroffen werden”, sagte Maier.

Two-stage: “Die Entscheidungen werden dann an anderer Stelle getroffen werden”, sagte Maier.

Idiom-only: “Die Entscheidungen werden dann an anderer Stelle getroffen werden”, sagte Maier.

LoRA r=16: “Dann werden Entscheidungen anderswo getroffen”, sagte Maier.

COMET: 0.936 / 0.942 / 0.942 / 0.943

In this general-domain example, all models produce plausible translations, but differ in fluency and conciseness. The LoRA model yields a more natural formulation (“anderswo getroffen”), while the other outputs remain slightly redundant due to repeated auxiliary constructions. Although these differences are subtle, they affect readability and are reflected in the small but consistent differences in COMET scores.